

7421

Integrated M.C.A. (First Semester)

Examination, Dec. 2022

IMCA-101

**PROGRAMMING FUNDAMENTALS
USING C**

Time Allowed : Three Hours

Maximum Marks : 70

Min. Pass Marks : 28

*This question paper contains three
sections as under :*

Section-A

Max. Marks-10

*This section shall have **one** compulsory question comprising **10** questions (maximum **20** words answer) taking **two** questions from each unit. Each question shall be of **one** mark and total marks of this section will be **10**. This section will be compulsory in the paper.*

Section-B

Max. Marks-25

This section will carry **25** marks with equally divided into five long answer type questions (answer in about **250** words) and students are instructed to attempt **five** questions by selecting one question from each unit.

Section-C

Max. Marks-35

Will carry **35** marks with **five** long answer type questions comprising one compulsory question (Question No. 7) of **15** marks and four questions of **10** marks each. Students are instructed to attempt total **three** questions with **one** compulsory question (answer about in **500** words) and any **two** more questions (answer about in **400** words) out of remaining four questions.

Section A

1. ~~Q~~ What are token in C Language ?

~~(i)~~ Define Algorithm.

~~(iii)~~ Explain Ternary Operator.

~~(iv)~~ Define Library Function.

~~(v)~~ Write a while loop to enter 10 numbers in array.

~~(vi)~~ Explain purpose of continue statement.

~~(vii)~~ Write the syntax to initialize 3D array.

~~(viii)~~ What is time complexity of Bubble Sort ?

~~(ix)~~ Define gets() function in C.

~~(x)~~ What is Union in C ?

Section B

UNIT-I

~~Q~~ Explain various datatypes in C Language.

Or

Write an algorithm to find least number in array of 10 numbers.

UNIT-II

3. Write a program to calculate sum and average of 10 numbers.

Or

4. Explain Bitwise operators in detail.

UNIT-III

4. Differentiate between if-else and switch in C Language.

Or

- Write a program to print the following pattern in C.

```
*
*  *
*  *  *
*  *  *  *
*  *  *  *  *
```

UNIT-IV

5. Write a program in C using recursion to calculate factorial of a number.

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[Contd....]

Or

- Explain the difference between call by ^{Pointer} value and call by reference argument passing.

UNIT-V

6. A book information contains the following fields :
Isbn, Book name, Book author, Book price and number of pages. Write a program using structure in C to enter and display record of 10 books with all above required fields.

Or

Explain the purpose of fopen() function and various file opening modes associated with it.

Section C

7. (a) Explain various types of arrays in C Language. ^{Name & Drived}
(b) Write an add function for adding two 3x3 array and display the result.

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[Contd....]

8. What do you understand by program design ?
Explain various techniques to represent the design of the program in C Language.
9. Explain nested control structure with suitable program code.
10. (a) Write a swap function using pointers to swap two numbers.
- (b) What are pointers in C ? Explain.
11. Write a program to copy the contents of file record1.C to record2.C.